<u>A Dissertation on The effect of security on practicality.</u>
<u>"How secured can a PC be before it is impractical?"</u>

Research doc..

<u>Dissertation Plan:</u>

- Discuss and Explore Software and Hardware vulnerabilities that a Personal Computer might encounter, possible flaws to common flaws.
- Discuss and Explore solutions in terms of how much they might impact system performance or interfere with user activity.
- Design an implementation of a highly secured and practical modern computer system.

<u>Problems insecure computers face:</u>

Software Vulnerabilities Computers may encounter:

<u>Malware :</u>

- ***<u>Types:</u>***
- Viruses : (code written by a bad actor to run in the background),
- Trojans : (Pretend to be one piece of software, but typically act with another untended purpose),
- Worms : (Viruses designed to spread, across a computer or a network)

<u>OS LEVEL Vulns:</u>

- Unpatched Security Flaws
- Insecure Default Settings
- Misconfigured User Accounts
- Weak Authentication and Authorization
- File System Vulnerabilities
- Lack of Address Space Layout Randomization (ASLR)
- Denial-of-Service (DoS) Vulnerabilities
- Insecure Logging and Monitoring
- Missing or Inadequate Intrusion Detection/Prevention Systems (IDS/IPS)
- Time-of-Check to Time-of-Use (TOCTOU) Vulnerabilities
- Zero-Day Exploits

<u>Hardware Vulnerabilities Computers may encounter:</u>

Dissertation Plan

Introduction

Provide an overview of the topic and its significance in modern computing.
Introduce the research question: "How secured can a PC be before it is impractical?"
Briefly outline the structure of the dissertation.

Literature Review

Define and explore the concepts of security and practicality in the context of personal computers.
Discuss previous research on the trade-off between security measures and user-friendliness.
Highlight relevant studies and findings related to software and hardware vulnerabilities.
Software Vulnerabilities in Personal Computers

Malware

Define and categorize different types of malware (viruses, trojans, worms).
Provide historical examples (e.g., Creeper & Reeper, Spanksa, Blaster) to illustrate the evolution and impact of malware.
Discuss common methods of malware propagation and exploitation.
Hardware Vulnerabilities in Personal Computers

Backdoors

Define hardware backdoors and discuss potential sources or instances.
Explain the implications of hardware backdoors on system security and user privacy.
Firmware Vulnerabilities
Detail potential vulnerabilities in BIOS/UEFI and peripheral firmware.
Provide examples of notable firmware vulnerabilities.
Impact Assessment of Security Measures

Discuss and evaluate various security solutions, both in terms of software and hardware.

Analyze the potential trade-offs between heightened security and system performance or user convenience.
Present case studies or experiments illustrating the effects of security measures on practicality.
Design and Implementation of a Highly Secured Computer System

Propose a model for a highly secured personal computer system.

Detail the components, software configurations, and hardware specifications necessary for implementation.
Discuss the anticipated benefits and potential challenges of the proposed system.
Conclusion

Summarize the key findings and insights from the research.

Reflect on the implications of balancing security with practicality in personal computing.
Suggest future areas of research or potential advancements in computer security.

References

Cite all the sources, studies, and materials used in the dissertation.

Aims of the dissertation

To de-stigmatise the difficulties and complexities concerning cyber security, preventative measures, and the maintenance of protection for the general public.

To provide an outline for, and improve the process of, building a secure system for domestic users, alongside users in an organisation.

To facilitate a system for implantation.

Objectives of the dissertation

To understand security, what makes a secure system and what makes a system secure.

To research and understand industry, military and domestic standards.

To provide instructions on how to build and configure a secure system.

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